

ISC Class XII Maths — Chapter 4

Determinants — Formula Sheet

A. Determinant Basics

$$2 \times 2: \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$3 \times 3 \text{ (expand along } R_1): \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Minor M_{ij} : determinant left after deleting row i , column j . **Co-factor** $A_{ij} = (-1)^{i+j}M_{ij}$

Expansion by cofactors (any row/column): $\Delta = a_{i1}A_{i1} + a_{i2}A_{i2} + a_{i3}A_{i3}$

B. Properties — Row/Column Operations

Swap $R_i \leftrightarrow R_j$ (or $C_i \leftrightarrow C_j$): determinant **changes sign**.

$R_i \rightarrow R_i + kR_j$ (or column analog): determinant **unchanged**.

Two identical/proportional rows or columns $\Rightarrow \Delta = 0$.

Common factor k in one row/column can be taken outside:

$$\begin{vmatrix} ka_1 & b_1 & c_1 \\ ka_2 & b_2 & c_2 \\ ka_3 & b_3 & c_3 \end{vmatrix} = k \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

C. Properties — Algebraic

$$|A^T| = |A| \quad |AB| = |A||B| \quad |A^n| = |A|^n$$

$$|kA| = k^n|A| \text{ for } n \times n \text{ matrix } A \text{ (not } k|A| \text{ — every row scales)}$$

$$|I| = 1$$

Tip: $|kA| = k^n|A|$ scales by k in *every* row, so the power is n (matrix size), e.g. $|2A| = 2^3|A| = 8|A|$ for a 3×3 matrix. Confusing this with $k|A|$ is the most common slip in this chapter.

D. Standard Determinant Identities

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \text{ -type with } a^3 \text{ swapped in for } a^2 \text{ column: } (a-b)(b-c)(c-a)(ab+bc+ca)$$

$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Worked Pattern: Row/Column Sum Trick

When every row (or column) shares a common sum (e.g. each row adds to $a+b+c$):

1. Apply $C_1 \rightarrow C_1 + C_2 + C_3$ (or $R_1 \rightarrow R_1 + R_2 + R_3$) to pull that common sum into one column/row.

2. Take the common factor outside the determinant.

3. Simplify remaining rows/columns with further $R_i \rightarrow R_i - R_j$ operations — zeros usually appear, making expansion easy.

E. Area of Triangle & Collinearity

Area of triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Take the **absolute value** (area can't be negative).

Collinear $\iff$ this determinant = 0 (area = 0).

Tip: **Equation of line** through $(x_1, y_1), (x_2, y_2)$: any point (x, y)

on the line satisfies $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$ — expand along R_1 to get the line's equation directly.

F. Singular Matrices & Adjoint

A is **singular** $\iff |A| = 0$ (non-singular $\iff |A| \neq 0$).

$\text{adj } A = C'$ where $C = [C_{ij}]$ is the cofactor matrix (transpose of cofactors).

2×2 **shortcut:** for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\text{adj } A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ — swap diagonal, negate off-diagonal.

G. Key Adjoint / Inverse Identities

$$A(\text{adj } A) = (\text{adj } A)A = |A|I \quad A^{-1} = \frac{1}{|A|} \text{adj } A \text{ (needs } |A| \neq 0)$$

$$|\text{adj } A| = |A|^{n-1} \text{ for } n \times n \text{ (so } = |A|^2 \text{ when } n = 3) \quad |A^{-1}| = \frac{1}{|A|}$$

$$\text{adj}(\text{adj } A) = |A|^{n-2}A \text{ for } n \times n$$

H. Inverse Properties & Reversal Laws

$$AA^{-1} = A^{-1}A = I \quad (A^{-1})^{-1} = A \quad (A^T)^{-1} = (A^{-1})^T$$

Reversal laws: $(AB)^{-1} = B^{-1}A^{-1} \quad \text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

If $AB = kI$ then $A^{-1} = \frac{1}{k}B$.

Worked Pattern: Inverse from a Matrix Polynomial

Given $A^2 - c_1A - c_0I = O$ with $c_0 \neq 0$:

- Rearrange: $A^2 - c_1A = c_0I$.
- Factor out A : $A(A - c_1I) = c_0I$.
- Divide by c_0 : $A^{-1} = \frac{1}{c_0}(A - c_1I)$.

Same idea for a cubic relation $A^3 - c_2A^2 - c_1A - c_0I = O$:

$$A^{-1} = \frac{1}{c_0}(A^2 - c_2A - c_1I).$$

Worked Pattern: Solving $AX = B$ (Martin's Rule)

- Write the system as $AX = B$ (coefficient matrix $\times$ variable column = constant column).
- Compute $|A|$; if $|A| \neq 0$, find $\text{adj } A$.
- $X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$.
- Read off each variable from the resulting column matrix X .

Tip: If the equations contain $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ instead of x, y, z : substitute $u = \frac{1}{x}, v = \frac{1}{y}, w = \frac{1}{z}$ first to make the system linear in u, v, w , solve $AU = B$ as above, then **back-substitute** $x = \frac{1}{u}, y = \frac{1}{v}, z = \frac{1}{w}$ — don't leave the answer in terms of u, v, w .

I. Consistency of $AX = B$

- $|A| \neq 0 \Rightarrow$ **unique solution** (consistent), $X = A^{-1}B$.
- $|A| = 0$ and $(\text{adj } A)B \neq O \Rightarrow$ **no solution** (inconsistent).
- $|A| = 0$ and $(\text{adj } A)B = O \Rightarrow$ **infinitely many solutions**.

J. Homogeneous Systems $AX = O$

- $|A| \neq 0 \Rightarrow$ **only the trivial solution** $x = y = z = 0$.
- $|A| = 0 \Rightarrow$ **infinitely many non-trivial solutions** (always consistent — $X = O$ is always one solution).

K. Common Traps to Avoid

- $|kA| = k^n|A|$, **not** $k|A|$ — every row is scaled by k .
- Swapping two rows/columns flips the *sign* of Δ — easy to forget mid-proof.
- A^{-1} exists only when $|A| \neq 0$ — always state this before writing $A^{-1} = \frac{1}{|A|} \text{adj } A$.
- For $|A| = 0$: don't assume "infinitely many solutions" automatically — check $(\text{adj } A)B$ first; it could be inconsistent instead.
- A homogeneous system is *never* inconsistent — $|A| = 0$ always gives infinite (non-trivial) solutions, never "no solution."

L. Quick Recognition Guide

Pattern seen	Use
Plain $2 \times 2 / 3 \times 3$ evaluate	Sec. A expansion
Determinant with a, b, c symmetric, rows share a sum	Worked Pattern: Row/Col Sum Trick
Prove $\Delta =$ product like $(a-b)(b-c)(c-a)$	Sec. D standard identity
"Find area" / "show collinear"	Sec. E
"Find equation of line through two points"	Sec. E line-through-two-points determinant
$ A = 0$ given/asked	Sec. F singular
"Find $\text{adj } A$ "	Sec. F cofactor/transpose
"Find A^{-1} " directly	Sec. G, $A^{-1} = \frac{1}{ A } \text{adj } A$
Given $A^2 =$ linear combo of A, I ; find A^{-1}	Worked Pattern: Inverse from Polynomial
"Solve using matrices" with 3 linear equations	Worked Pattern: $AX = B$
"Check if consistent"	Sec. I — test $ A $, then $(\text{adj } A)B$
System = 0 on the right (homogeneous)	Sec. J trivial/non-trivial
$(AB)^{-1}$, $\text{adj}(AB)$ in a proof	Sec. H reversal laws